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ORIGIN OF HEMATITE AND ITS ASSOCIATED REMANENCE IN THE COPPER HARBOR CONGLOMERATE (KEWEENAWAN), UPPER MICHIGAN

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Abstract. To study further the origin remanence in red beds, the volcanogenic Copper Harbor Conglomerate has been investigated using both paleomagnetic and petrographic techniques. Paleomagnetic results suggest the presence of two magnetic components, both possessing a westerly and shallow or normal Keweenawan direction. One component (in the vicinity of declination 294°, inclination -7.6) comprises approximately 30% of the total natural remanent magnetization, is removed by alternating field demagnetization following chemical demagnetization, and shows a consistent inclination error that is controlled in sense and amount by the dip of the cross beds in which the samples were taken. The low coercivities of this component suggest that the magnetization resides in magnetite. This and the inclination error due to the bedding indicate that the component is a detrital remanent magnetization.

The predominant component (declination 286.5°, inclination 0.0°, based on six sites) is removed by chemical demagnetization and shows no consistent relationship between inclination and dip of the cross beds, which suggests that it is a postdepositional magnetization. This component resides in hematite that is removed during the chemical demagnetization procedure. Petrographic studies show that hematite is present authigenically as pigment and as partly to wholly martitized grains. Martitization may have occurred before or after deposition; petrographic results are at present inconclusive. A pole position calculated from this postdepositional component (168.3°E, 11.2°N) is similar to poles from the overlying Nonesuch Shale and Freda Sandstone, which suggests that the remanence was acquired before the end of Freda deposition.

Introduction

The genesis of hematite pigment in red beds has long been the subject of considerable debate. For example, Krynine [1949] thought that the pigment was detrital, directly eroded from red tropical soils (latosoils). Van Houten [1964] believed that the pigment was diagenetic, having formed in situ by aging of detrital yellow or brown iron oxides. Walker [1967], however, persuasively argued that most pigment is authigenic, having formed diagenetically over long periods of time by in situ alteration of detrital silicate grains. At the present time the most widely accepted hypotheses are the following two: in situ aging of a detrital ferric oxide precursor [e.g., Friend, 1966; Braunagel and Stanley, 1977; Hubert and Reed, 1978; McPherson, 1980] and diagenetic alteration of silicate grains [e.g., Schluger, 1976; Walker, 1976; Walker et al., 1981].

This debate has recently been rekindled, particularly as it relates to the origin of magnetization in red beds, which usually resides in hematite. Although red beds have been extensively used in paleomagnetic investigations, the origin of the remanence in such deposits is still poorly understood, and the subject is one of considerable controversy. Some investigators believe that hematite in red beds can be acquired relatively early and that its remanance accurately reflects the geomagnetic field during deposition. Consequently, the remanence can be used in magnetostratographic studies and to determine relatively well-dated paleopoles [Elston and Purucker, 1979; Purucker et al., 1980]. Other workers, however, impressed by secondary magnetizations caused by diagenetic hematite, question the use of red beds in such studies [Turner, 1979, 1980; Walker et al., 1981]. The time of remanence acquisition is the crucial issue in this debate. To examine further this issue and the general problem of the origin of hematite in red beds, we have undertaken a combined paleomagnetic, rock magnetic, and petrographic study of a red bed deposit, the Keweenawan Copper Harbor Conglomerate.

Because many red beds contain more than one stable component of magnetization [Roy and Park, 1972; Turner and Ixer, 1977; Van der Voo et al., 1978; Henry, 1979; and numerous others] and more than one type of hematite [e.g., Walker et al., 1981], an important objective in the study of such beds is isolation of the different components and identification of their magnetic carriers. In particular, distinguishing a detrital remanent magnetization (DRM) from any secondary or chemical remanent magnetizations (CRM's) has obvious implications in a paleomagnetic study. Studies of DRM have demonstrated that it can be complex, affected by a number of depositional and postdepositional processes that can cause deviations in the remanent magnetization [Verosub, 1977]. These deviations include shallowing of inclinations in horizontal beds ('inclination' error; King [1955]), differences in inclination controlled by the dip of cross bed laminations (bedding error; King [1955]), declination and inclination changes due to flowing water [King, 1955; Granar, 1958], and changes in direction due to postdepositional rotation of grains (postdepositional DRM; Irving [1957] and Kent [1973]). In this study we have used one of these effects, bedding error, in conjunction with a petrographic investigation, to identify a DRM and

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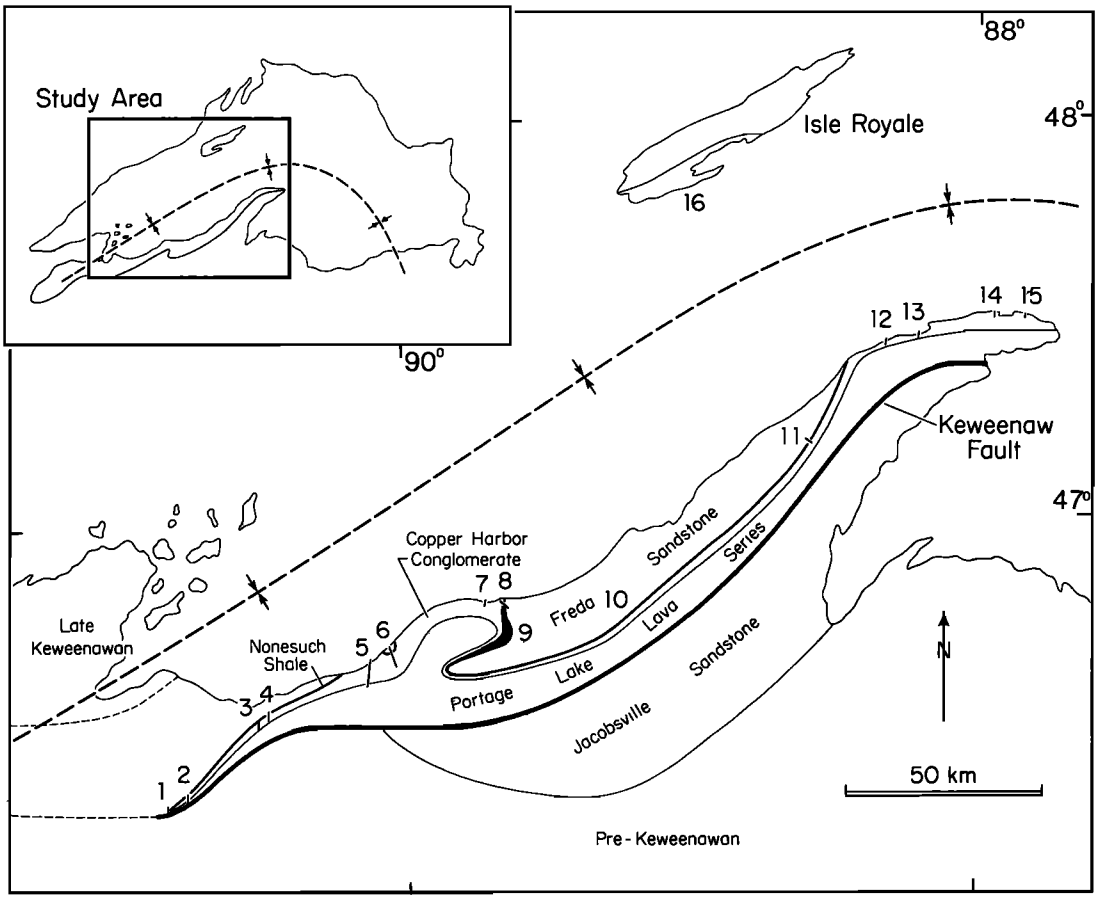


Fig. 1. Index and geologic map of the area surrounding the sampling location. Axis of Keweenawan Trough is shown on both maps. Numbers represent sections studied [Elmore, 1981]; sampling for this study was at location 7.

distinguish it from any CRM's. In this way we have been able to understand better the origin of hematite and its associated remanence in one red bed unit.

General Geology

The Copper Harbor Conglomerate is part of a thick sequence of clastics and volcanics that fills the Keweenawan Trough in the Lake Superior region of North America (Figure 1). On the southeast limb of the trough in northern Michigan the sequence begins in the Portage Lake Lava Series, which is composed of mafic lavas and subordinate felsic clastics [White, 1968]. The volcanics are overlain by the Copper Harbor Conglomerate, a fining-upward wedge of volcanogenic conglomerate and sandstone deposited as proximal-to-distal braided stream and sheetflood facies on coalesced alluvial fans and sandflats [Elmore, 1981]. The Copper Harbor Conglomerate is conformably overlain by the black-to-green lacustrine silts and shales of the Nonesuch Shale, which is in turn overlain by the red, fluvial sandstones and shales of the Freda Sandstone (Figure 1).

Samples were collected from one exposure of the upper Copper Harbor Conglomerate along the Lake Superior shoreline (Figure 1, section 7, 46°49'N, 89°38'W). At this site the Copper Harbor Conglomerate dips 20°N and consists predominantly of red festooned or trough cross-stratified sandstone. Other abundant rock types include current-rippled sandstone, wave-rippled sandstone, and horizontally laminated sandstone with current lineations. Mud drapes, mud intraclasts, and mud cracks are ubiquitous in this facies. Current directions determined from a variety of structures indicate flow to the northeast. This facies represents deposition by braided stream and sheetflood processes on the distal part of an alluvial fan [Elmore, 1981].

The age of the Copper Harbor Conglomerate is bracketed between 1000 and 1100 Ma. The upper limit is the age generally assigned to the Portage Lake Lava Series [e.g., Irving and McGlynn, 1976], while the lower date is based on a Rb-Sr date of 1007±25 Ma from a rhyolitic plug intruded into the Freda Sandstone (S. Chaudhuri, personal communication from Henry et al. [1977]).

Sampling and Laboratory Analysis

All samples were collected with a portable coring apparatus and oriented with a Brunton compass and clinometer. Eighty cores were collected and investigated from 11 closely spaced sites in cross beds of different attitudes. The sites were selected from as large a range of cross bed directions and dips as possible

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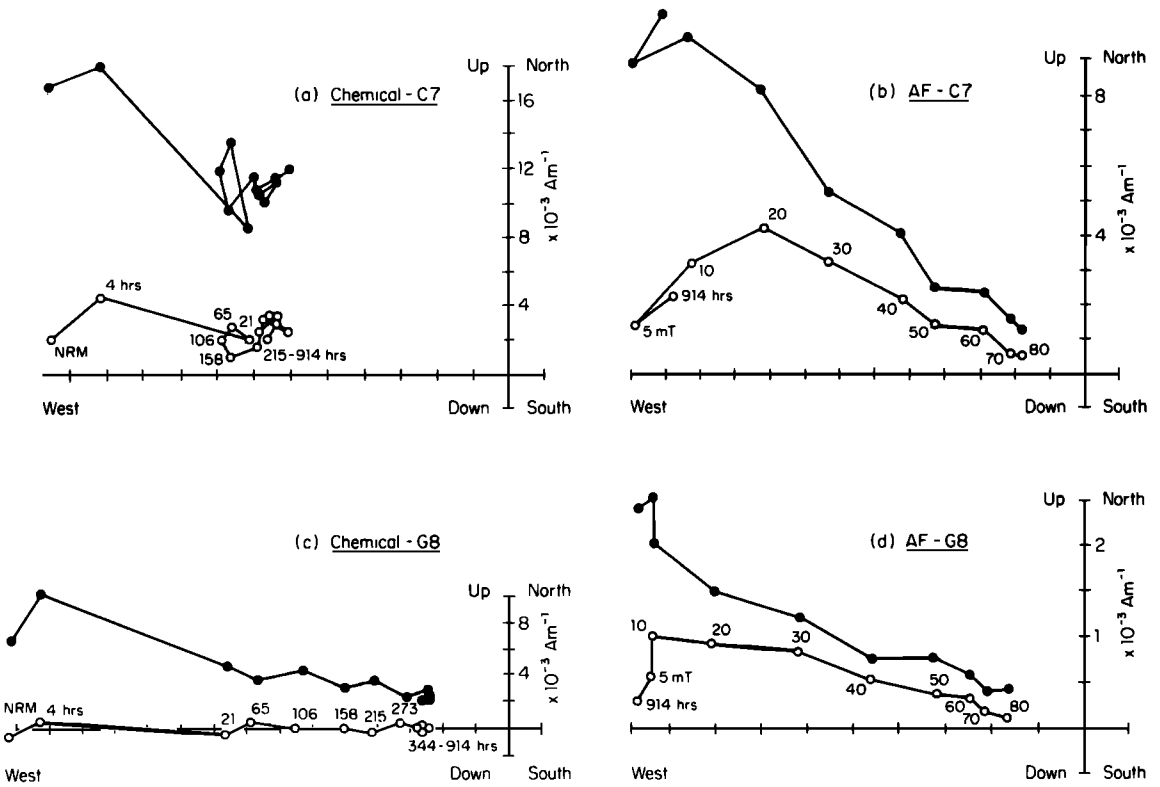


Fig. 2. Orthogonal projections [Zijderveld, 1967] of the successive end points of the magnetization vector during chemical demagnetization (Figures 2a and 2c) followed by AF demagnetization (Figures 2b and 2d) for two samples. Solid symbols represent projections on the horizontal plane and open symbols on the vertical plane after a structural correction of 20°.

in order to test for a bedding error. Because of outcrop bias, however, most sites were in southeasterly dipping cross beds. All samples were collected from a stratigraphic interval of approximately 10 m.

Standard cylinders of 2.2-cm height were cut from 2.54-cm-diameter cores, and their natural remanent magnetization (NRM) measured with a ScT cryogenic magnetometer. A few specimens were thermally demagnetized using a Schonstedt TSD-1 unit, although most specimens were cleaned using chemical followed by alternating field (AF) demagnetization. Chemical demagnitization was carried out by cutting three vertical slots in the specimens and then soaking in 10 N HCL in field-free space for set periods of time before each measurement. The specimens were dried in field-free space prior to measurement. As the procedure progressed and had less effect, treatment times were increased. Most specimens were leached for 500-900 cumulative hours and a few for 1800 hours, although treatment longer than 400 hours did not reduce the magnetization significantly. Typical demagnetization diagrams for chemical treatment are shown in Figure 2. At different stages of the treatment, some specimens were periodically removed for petrographic examination.

Chemical demagnetization typically removed from 50 to 80% of total NRM. After leaching, the remaining remanence was removed by AF demagnetization in fields from 5-80 mT, using a Schonstedt GSD-1 unit. Vector subtraction [Zijderveld, 1967, 1975] was used to determine the directions of the components removed during demagnetization. Isothermal remanent magnetization (IRM) acquisition curves were determined for a few leached specimens with a cryogenic magnet capable of reaching 6 T.

Polished thin sections of untreated specimens and of specimens removed during stages of the leaching process were studied under both transmitted and reflected light. In addition, representative samples were examined using a scanning electron microscope (SEM) with attached Kevex X ray energy spectrometer (XES).

Paleomagnetic Results

NRM intensities of specimens ranged from 1.5 to 9.1 x 10^{-2} A/m. The specimens could be divided into two groups based on their behavior during chemical demagnetization. In the generally coarse-grained sandstone specimens, leaching removed approximately 50% of the total NRM in the first 21 hours; thereafter, leaching had little effect on the remaining magnetization (Figure 2a). During subsequent AF demagnetization, however, the magnetic vector did decay to the origin, and a direction could be calculated with confidence (Figure 2b). Leaching typically removed 70-80% of the NRM in fine-grained sandstone specimens, and directions could be determined from both chemical and AF demagnetization diagrams (Figures 2c and 2d).

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TABLE 1a. Paleomagnetic Summary of Site-Mean Directions

Site	Cross Bed Dip, deg	n/Rsp	k	α95	Decl., deg	Incl., deg
A chem.	0	9/9	174	3.9	271.2	-3.0
AF		6/6	53	9.3	297.1	-7.6
B chem.	5SE	8/8	95	5.7	291.2	-7.8
AF		6/6	137	5.8	292.7	-18.3
C AF	12SE	7/7	57	8.1	301.3	-22.4
D AF	16SE	6/6	124	6.0	298.1	-26.5
E AF	4SE	3/3	1284	3.4	293.0	-15.3
F chem.	11SE	5/5	184	5.7	290.4	-2.0
AF		3/3	1292	3.4	294.7	-19.0
G chem.	1SE	6/6	209	4.6	280.5	1.0
AF		6/6	321	3.7	296.4	-11.2
H chem.	12SE	8/8	92	5.8	280.1	2.8
AF		8/8	89	5.9	289.1	-20.0
I AF	2SE	5/5	81	8.5	278.0	-14.3
J chem.	12NW	2/2	(40)	(40.6)	296.0	5.6
AF		2/2	(108)	(24.3)	282.5	0.5
K chem.	.5SE	7/7	166	4.7	296.7	3.6
AF		7/7	178	4.5	292.4	-9.8

Chem., component removed by chemical demagnetization and AF, component removed by AF demagnetization. Cross bed dip is corrected to that parallel to a direction of N290°E; n/Rsp is the ratio of specimens measured to specimens used for statistical analysis; k and α95 are statistical parameters [Fisher, 1953]; Decl., declination and Incl., inclination.

Two major magnetic components have been identified from most specimens; one removed by chemical treatment (mean declination of 286.5°, inclination 0.0°) and a low coercivity component (site A: declination 294°, inclination -7.6°) removed by AF demagnetization after chemical leaching (e.g., Figure 2). Both components possess a westerly and shallow direction, with the component removed by chemical demagnetization having a consistently more positive inclination than the component removed by AF demagnetization (Figure 2). In addition to these major components, a northwesterly and steeply down direction, thought to be in part of recent and viscous origin, is removed in alternating fields up to 30 mT (Figures 2b and 2d). Thermal demagnetization did not separate the two major components but showed univectorial decay to the origin giving a westerly and shallow direction. The site mean directions and associated statistics are summarized in Table 1, with unit weight given to specimens.

The specimen directions for the component removed by AF demagnetization are shown in Figure 3. Figure 4 illustrates that the differences in inclination are controlled by the dip of bedding surface; there is a systematic relationship between the bedding error (β) or the difference in inclination between horizontal and dipping beds, and the dip of the bedding surface (α). Bedding error (β) was calculated by taking the difference in inclination between the horizontal bed (after structural correction) and each of the inclined cross beds. Site A was the only site in a horizontal bed, so it was used as the reference DRM site. The test for bedding error is therefore positive; the inclinations are controlled by the sense and magnitude of the dip of the cross beds.

Figure 3 also shows the specimen directions for the component removed by chemical demagnetization. The samples generally plot as a large cloud with deviations in declination as large, or larger than, the deviations in inclination. As illustrated in Figure 5, there is no systematic relationship between bedding error and dip

TABLE 1b. Summary: Pole Positions

	n/Rs	Decl. deg	Incl. deg	k	α95	Pole Position Longitude E, deg	Pole Position Latitude N, deg	dp	dm
Chem., giving unit weight to site means	7/7	286.5	0.0	58.0	8.0	168.3	11.2	4.0	8.2
AF (site A), giving unit weight to specimens	6/6	297	-7.6	53.0	9.3	157.5	15.3	4.7	9.6

The dp and dm are the semiaxes of the ovals of 95% confidence around the poles.

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of the bedding surface, the component removed by chemical demagnetization failed the bedding error test.

Thermal demagnetization curves show that the specimens lose their magnetization through a range of temperatures (Figure 6), indicating a wide distribution of blocking temperatures [Irving and Opdyke, 1964]. Collinson [1974] has shown that in red beds where hematite has a narrow range of blocking temperatures, specularite is the dominant carrier of remanence, whereas in examples where hematite has a broad range of blocking temperatures, pigment is the dominant remanence carrier. Although Collinson's [1974] technique of comparing the thermal demagnetization of separated pigment and specularite/martite with the whole rock NRM was not performed, the wide range of blocking temperatures in the Copper Harbor Conglomerate samples suggests that most of the remanence resides in pigment.

Figure 7 shows a typical IRM acquisition curve for a leached sample. The IRM reaches 70-80% of saturation in magnetizing fields of 0.3 T and then gradually rises to over 3T. These results suggest the presence of both a low coercivity mineral such as magnetite and a high coercivity phase, probably hematite. The specimens were still slightly red due to hematite that was not leached, which would explain the presence of the high coercivity phase. The generally low coercivities indicated by AF demagnetization of leached specimens, however, suggest that magnetite is the major carrier of this component.

Petrographic Results

Two potentially magnetic minerals were identified in the samples: magnetite and hematite.

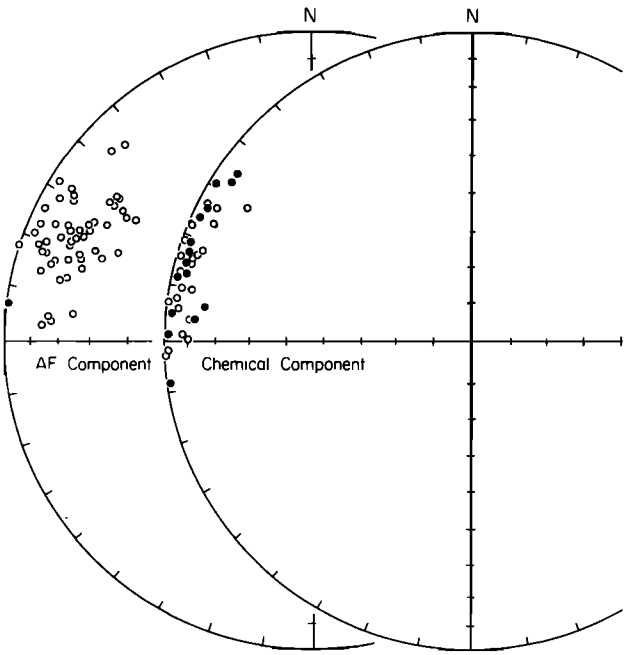


Fig. 3. Equal-area projections of the components removed by AF and chemical demagnetization. Specimens plotted after structural correction but without correction for foreset dip. Open symbols pointing upward, solid symbols pointing downward.

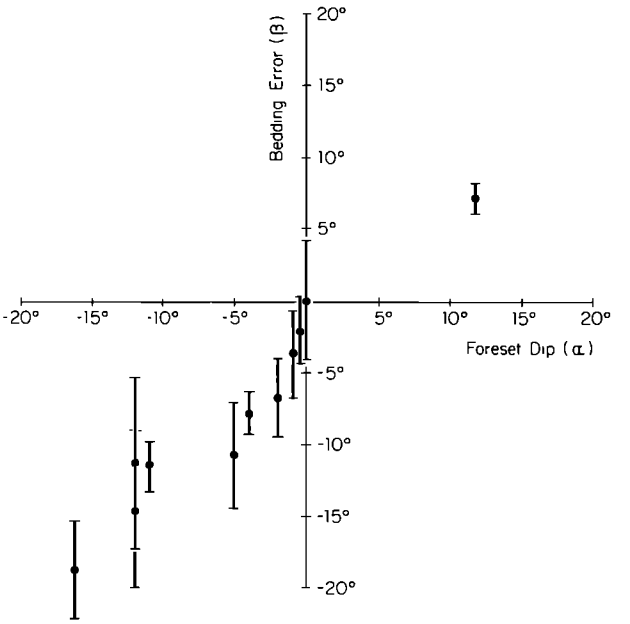


Fig. 4. Graph of bedding error (β) versus the apparent dip of the bedding plane (α) for each site for the component removed by AF demagnetization; the apparent dip is the dip corrected to that in a plane parallel to the Keweenawan declination of 290°E. All points in the lower left quadrant come from southeasterly dipping beds, while the one point in the upper right comes from a northwesterly dipping bed. Error bars are the standard deviations of the inclinations for each site. Site A (horizontal bed) is set at 0°. All cross bed dips after structural correction. The correlation coefficient for the data is 0.97.

Magnetite, identified in reflected light under oil immersion, is found as relicts in partly martitized grains (Figure 8a). Many grains which appeared to be homogeneous specularite or martite at low magnifications were found to contain the magnetite relicts at magnifications greater than 500X.

Hematite occurs as pigment and as discrete oxide grains. Most of the red color in the Copper Harbor Conglomerate is imparted by a stain (ultrafine pigment of Walker et al., [1981]) that irregularly coats grains and is usually associated with clay rims. Most of these clay rims have a boxwork type of texture, which indicates that they are authigenic (Figure 8b). The hematite associated with such rims is probably also authigenic. Other rims are composed of clay platelets that are aligned parallel to the grain surfaces and are often found at grain-to-grain contacts, which suggests that they coated the grains before deposition. The hematite associated with these rims may have formed by dehydration of ferric oxide precursors which stained the clay before deposition. In any case, the stain on both types of clay rims is either amorphous or too small to be identified with the SEM. Although undetermined, the grain size is probably below the super paramagnetic size limit for hematite, in which case the pigment probably does not contribute greatly to the remanence. Euhedral crystals approximately 1 μm in size are also found on grain surfaces

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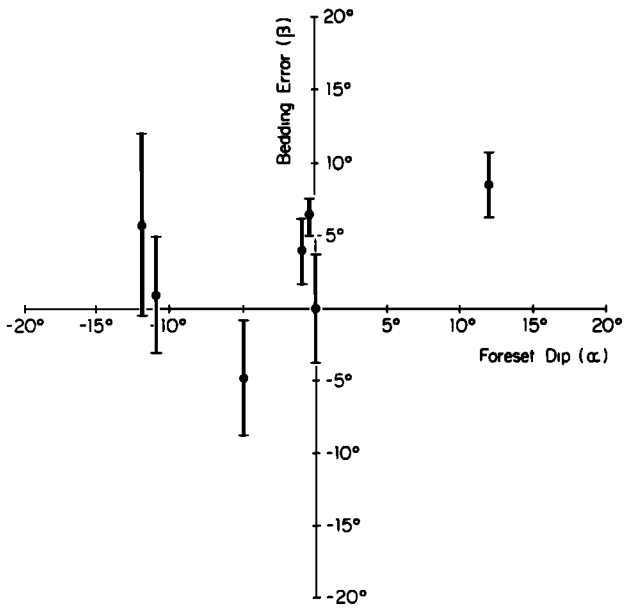


Fig. 5. Graph of bedding error (β) versus apparent dip of the bedding plane (α) for the component removed by chemical demagnetization. The conventions are the same as in Figure 4.

(Figure 8c) and are coarse enough to carry a stable remanence [Banerjee, 1971]. They are clearly authigenic.

Petrographic evidence suggests that a principle source of iron for the authigenic hematite was alteration of iron-bearing silicate grains. The Copper Harbor Conglomerate consists predominantly of debris derived from mafic-to-felsic volcanic rocks. This material has been altered by a variety of diagenetic changes other than the formation of authigenic hematite, including partial-to-complete dissolution of silicate framework grains, and precipitation of authigenic feldspar, quartz, clay, laumontite, sphene, and titanium oxides. Iron-bearing silicate grains are almost totally absent, which has important implications since such grains must have been common at deposition. Their dissolution released abundant iron ions which may have been the source for authigenic hematite [Walker, 1976]. In addition, some of the opaques are altered to anatase, which could have released additional iron ions.

Hematite of possibly detrital origin occurs in a number of textural varieties. These include partly to wholly martitized grains with triangular 'exsolution' lamellae of ilmenite or one of its alteration products (Figure 8a) and monocrystalline specularite. In addition to occurring as discrete grains, martite is also found in volcanic fragments (Figure 8d). The hematite in the partly martitized grains has replaced the magnetite from the lamellae inward and replacement is generally more pronounced on grain margins, often with uniform outer rinds of martite (Figure 8e).

Thin section observations and the red color of the acid during leaching indicate that both the authigenic and the possibly detrital hematite were removed during chemical demagnetization. It was hoped that leaching would remove the fine-grained, presumably more soluble authigenic hematite before the coarser, less soluble martite. This would have allowed identification of the components residing in each type [Henry, 1979]. Unfortunately, leaching was not uniform, with hematite removed in stages from the margins inward. Although the hematite in pigment form was removed slightly before detrital hematite, there was considerable overlap. The reason for this overlap is partly due to similar solubility spectra [Henry, 1979] but also to the nonuniform nature of leaching. The measured magnetization probably represented both the removed pigment and the remaining detrital hematite, as an averaged vector sum. If the two types of hematite did carry different magnetizations, they were not separated.

Discussion

The presence of a consistent bedding error controlled by cross bed dip in the component removed by AF demagnetization is probably best explained by rolling of grains during deposition [King, 1955; Griffiths et al., 1960], which indicates that it is a DRM. Removal of the component by AF demagnetization strongly suggests that the remanence resides in magnetite. This provides further evidence for a DRM. The lack of a consistent bedding error in the component removed by chemical demagnetization and removal of obviously authigenic hematite during the procedure suggests that it is a CRM. These conclusions have a number of important implications.

The fact that thermal demagnetization did not resolve the two components points out the value of employing multiple demagnetization techniques. This particularly relates to the recognition of a detrital remanence residing in magnetite. Although there have been some reports of relict magnetite in specularite [e.g., Turner and Ixer, 1977; Walker et al., 1981], identification of the component it carries requires careful demagnetization procedures. If the predominant remanence in red beds resides in secondary hematite, as is the case in this study and in many other investigations of red bed remanence [e.g., Turner, 1979], then identification of a DRM in magnetite has obvious implications.

It must be noted, however, that the magnetite

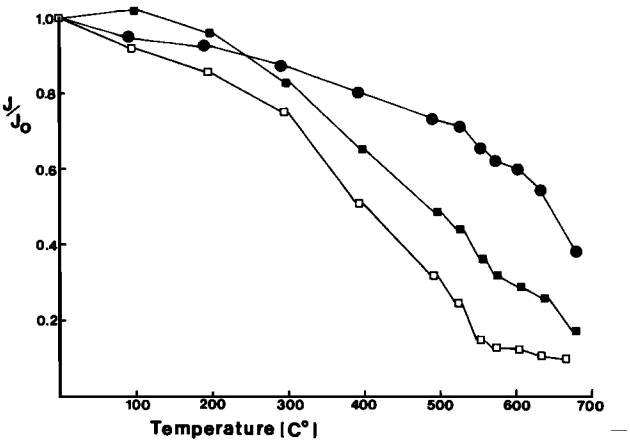


Fig. 6. Normalized thermal demagnetization curves for three samples. Solid circles, UBA-2; open squares, UBC-1; solid squares, UBB-2.

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in this study carries a VRM as well as a DRM, and that additional AF demagnetization is needed after chemical treatment to truly isolate the DRM. It is most likely that for any paleomagnetic study of red beds containing magnetite as well as hematite, chemical demagnetization alone will not be adequate.

The existence of a bedding error in cross-stratified sandstones, however, demonstrates that such beds should not be used for depositional pole calculations. There are problems even with horizontally laminated sandstones, since 'inclination' erros [King, 1955] are possible. Because of these problems with sandstones, some workers suggest that poles should be calculated from siltstone and claystone [Elston and Purucker, 1979]. Yet the magnetizations in these lithologies can undergo postdepositional changes and can also be of secondary origin. If such lithologies are to be used, then conglomerate and synsedimentary fold tests should be performed to determine if the remanence is contemporaneous with deposition [e.g., Purucker et al., 1980]. Although a DRM in sandstones may be of questionable value for depositional paleopole calculations, it accurately records polarity and can be used in magnetostratigraphic studies [Elston and Purucker, 1979].

Although our paleomagnetic evidence points to only one component residing in hematite, petrographic observations demonstrated the presence of abundant grains that are of probable detrital origin (predominantly martite) as well as authigenic hematite. This raises an important question: Were the grains deposited as martite or as magnetite that was later martitized? If martitization took place in the source area during deuteric 'oxyexsolution' (which caused the ilmenite lamellae) and pseudomorphic oxidation [Haggerty, 1976], then the martite could carry a DRM. On the other hand, if the martitization formed in situ and was diagenetic in origin, the grains would carry a CRM. This same analysis applies to the monocrystalline grains in the Copper Harbor Conglomerate, which could have formed from preexisting hematite or by replacement [Walker et al., 1981].

A number of workers believe that the martitization observed in ancient red beds is largely postdepositional and cite the following evidence: Eh-pH conditions in red beds favorable for alteration and abundant magnetite in many modern depositional settings with a depletion of magnetite relative to specularite in successively older rocks [Van Houten, 1968; Turner, 1979; Walker et al., 1981]. Application of this indirect evidence to the volcanogenic Copper Harbor Conglomerate, however, is not obvious. Deuteric pseudomorphic oxidation can clearly produce abundant hematite from magnetite in volcanic rocks [Haggerty, 1976]. In fact, much of the hematite in the Portage Lake Lava Series, which underlies the Copper Harbor Conglomerate, apparently formed during high-temperature deuteric oxidation [Butler and Burbank, 1929; Ade-Hall et al., 1971].

Definitive petrographic criteria for distinguishing between deuteric and low-temperature replacement have not been determined. Elston and Purucker [1979] have suggested that because specularite has a pseudocleavage that produces tabular grains, the presence of such grains on

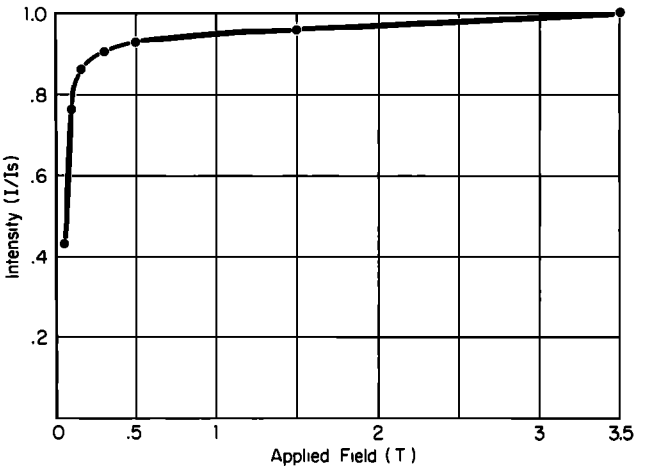


Fig. 7. Typical IRM acquisition curve for a leached specimen. Note the rapid rise of the magnetization up to 0.3 T and then the gradual rise from 0.3 T to over 3.5 T. This suggests the presence of both a high and a low coercivity phase.

bedding planes indicates that they were deposited as specularite. These types of grains were not observed in the Copper Harbor Conglomerate. The relatively uniform rinds of hematite around relict magnetite in some Copper Harbor Conglomerate grains suggest that such grains are diagenetic in origin; if the replacement had occurred in the source area, abrasion and breakage should have resulted in less uniform rinds.

In summary, the petrographic evidence is inconclusive: some grains were probably deposited as martite, with others martitized after deposition. This particular problem clearly merits additional work. The paleomagnetic evidence, however, suggests that most of the remanence is of secondary origin. In addition, the wide range of blocking temperatures suggests that most of the remanence resides in pigment. If some of the martite grains do carry a DRM, it is being masked by the stronger CRM residing in authigenic hematite, and chemical demagnetization did not separate the two components. The latter point emphasizes that although chemical demagnetization is a useful technique, if it is to be effective in separating components residing in hematite, great care must be taken to insure uniform leaching. Nonuniform leaching, which is probably the rule rather than the exception, can mask the different components.

Because most of the remanence in the Copper Harbor Conglomerate is secondary, it is of questionable value for determining depositional paleopoles and for studies of secular variation and contemporaneous polarity transitions. Some workers suggest, however, that even though a magnetization is not detrital, if it is acquired relatively soon after deposition, it may still be suitable for paleomagnetic analysis [Purucker et al., 1980]. Estimates of the time lag between deposition and acquisition of such magnetizations are approximately 10^5 years [Purucker et al., 1980; Channel et al., 1982]. Other workers, however, argue that although such early magnetizations can yield depositional

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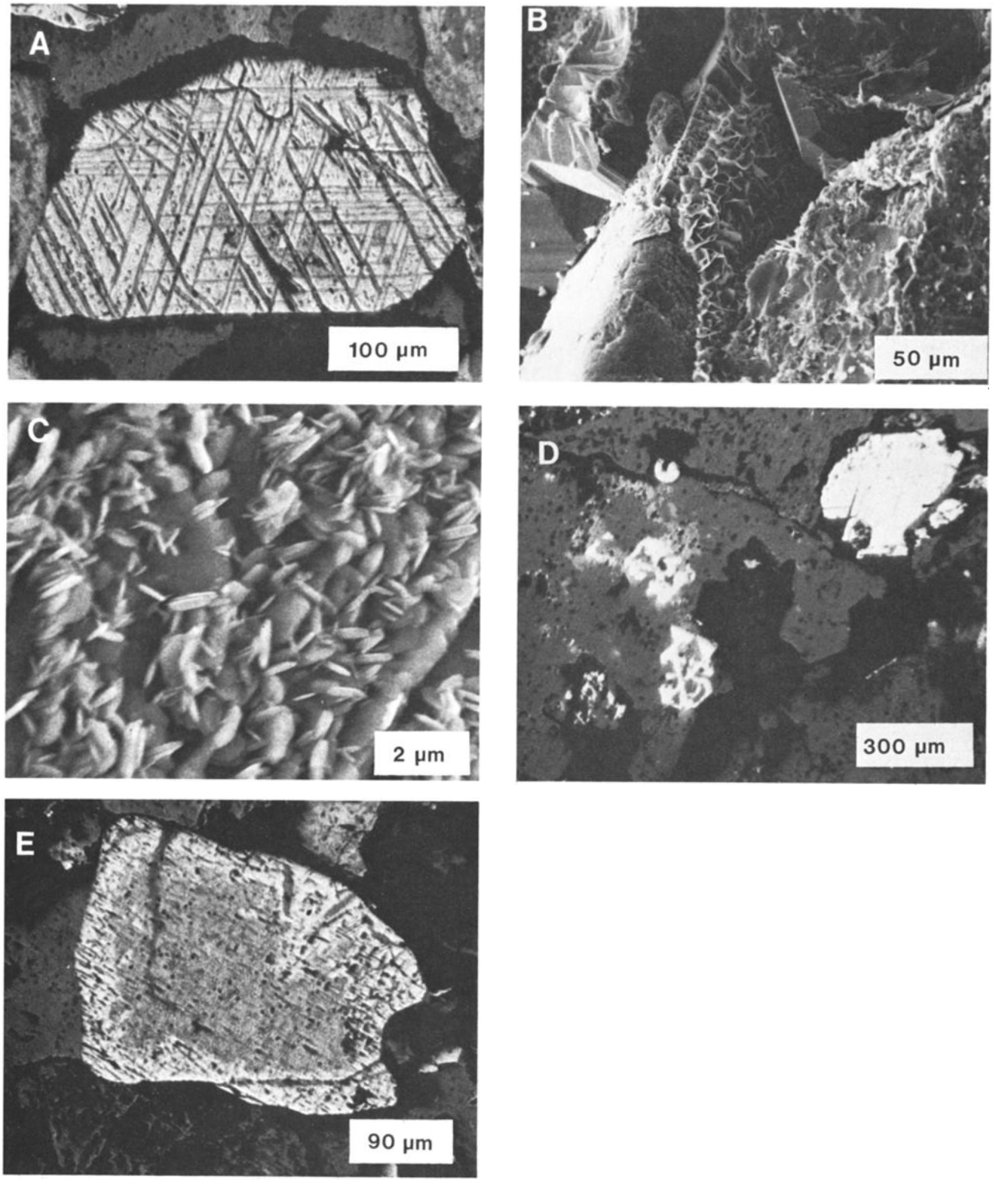


Fig. 8. Photomicrographs of detrital and authigenic minerals in the Copper Harbor Conglomerate. (a) Martite grain with exsolved lamellae of ilmenite or one of its alteration products with one gray patch that is relict magnetite. Reflected light. (b) SEM photomicrograph showing authigenic clay coatings on grains and a euhedral quartz overgrowth. (c) SEM photomicrograph showing euhedral crystals of authigenic hematite. (d) Rock fragment in which hematite has replaced cruciform-type magnetite. Reflected light. (e) Trellis type grain with relict magnetite (dark phase, in center of grain), lamellae of a titanium-rich phase, and an outer rim of hematite (light phase) that has replaced magnetite. The relatively uniform rim of hematite suggests that replacement was secondary. Reflected light.

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pole positions of sufficient precision, they are unsuitable for studies of polarity transitions and secular variation [Turner, 1979].

To examine further the critical issue of timing for the Copper Harbor Conglomerate, pole positions were calculated for both the DRM (site A; in a horizontal bed; 157.5°E, 15.3°N) and the CRM (168.3°E, 11.2°N) and compared to other poles in the late Keweenawan apparent pole wander path (APWP). The APWP shows an increase in paleopole distance from the present pole with decreasing age (Figure 9). The poles calculated from this study agree fairly well with Irving's [1979] late Keweenawan APWP, although they are offset to the west. This offset is problematic and may be due to secular variation. It should be noted, however, that the poles are similar to poles from late Keweenawan clastics reported by Watts [1981] that also require a modification of the late Keweenawan APWP. In addition, the poles coincide with the younger end of the Grenville Loop [Morris and Roy, 1977].

The DRM and CRM poles are similar. The difference in paleopoles may be attributable to apparent polar wander, with the DRM older than the CRM, or incomplete averaging of secular variation. It should be noted, however, that the DRM pole was only based on six specimens at one site, and as a result, it should not be considered a pole position. It was included in this study only for comparison with the CRM pole.

The CRM pole is most similar to a primary pole determined from the Nonesuch Shale (176.5°E, 10.6°N; Henry et al. [1977]), indicating that the remanence may have been acquired during deposition of that unit. Other secondary poles have been reported from conglomerate clasts in the Copper Harbor Conglomerate (179.3°E, 11.5°N; Palmer et al. [1981]) and from the Nonesuch Shale-Freda Sandstone (169.2°E, 3.7°N; Henry et al. [1977]). These poles and the CRM pole in this study are all similar, and the magnetizations that give rise to them were probably acquired before the end of Freda deposition. Although this is the favored interpretation, the coincidence of the CRM pole with the younger end of the Grenville Loop (900-950 Ma) [Morris and Roy, 1977] suggests the possibility that the CRM pole could be significantly younger than Freda deposition.

A Freda age suggests that the remanence was acquired relatively early although because of uncertainties about the pole positions and the lack of well-defined dates for the units, an absolute time cannot be determined. The time lag between deposition and acquisition of the secondary remanence, however, was probably greater than 10^5 years, which indicates that it is unsuitable for paleomagnetic analysis in the context of depositional age of the rocks. The magnetization does, however, have value in that it dates the formation of diagenetic hematite.

Conclusions

Petrographic results show that hematite in the Copper Harbor Conglomerate occurs in at least three forms: ultrafine pigment, microcrystalline hematite, and as partly to wholly martitized grains. Martitization may have occurred before or after deposition; petrographic results are at present inconclusive.

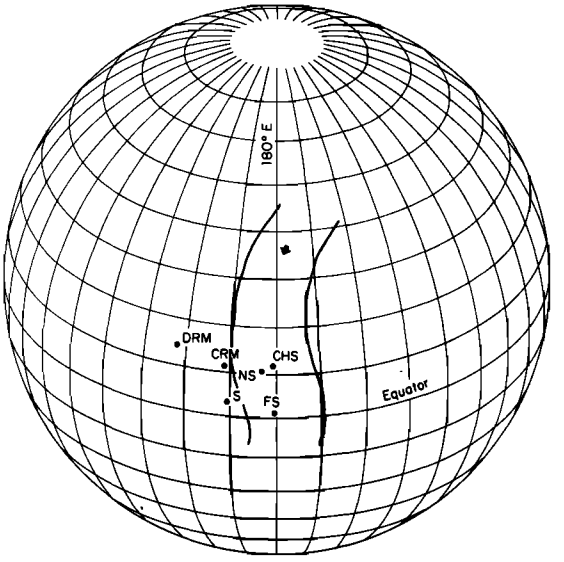


Fig. 9. Late Keweenawan apparent polar wander path for North America after Morris and Roy [1977]. The DRM and CRM poles are from this study. NS, Nonesuch Shale; FS, Freda Sandstone; S, Nonesuch-Freda secondary [Henry et al., 1977]; CHS, Copper Harbor Conglomerate secondary pole in conglomerate clasts [Palmer et al., 1981].

Most of the ultrafine and all of the microcrystalline hematite is authigenic, with postdepositional alteration of iron-bearing silicates the primary source of iron ions.

The paleomagnetic results of this study show that the Copper Harbor Conglomerate contains at least two magnetic components. The presence of a systematic bedding error and identification of magnetite as the predominant magnetic carrier of one component identifies it as a DRM. Because of the considerable inclination error in cross-bedded sandstones, such beds should not be used to determine depositional pole positions. A DRM can be used, however, in magnetostratigraphic studies.

The fact that the DRM was masked by a CRM and was only identified by AF demagnetization after chemical demagnetization points out the usefulness of such techniques in paleomagnetic studies of red beds. Red beds are often complex paleomagnetically, and only by careful demagnetization can the components be resolved.

Most of the remanence in the Copper Harbor Conglomerate resides in authigenic hematite and was probably acquired before deposition of the upper Freda Sandstone. The CRM pole position is similar to other secondary poles from the Copper Harbor Conglomerate and paleopoles from the Nonesuch Shale-Freda Sandstone. This remanence which resides in hematite is less valuable for paleomagnetic analysis if it is considered to be important that the age of the rock and the age of the magnetization must be the same. The magnetization does have value, however, in that it suggests that diagenetic hematite was acquired relatively early.

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